

ANINDYA SUNDAR NANDI

Software Development Engineer | Full-Stack | Backend Systems

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• Portfolio • LinkedIn • GitHub

PROFESSIONAL SUMMARY

Software Development Engineer with a strong computer science and mathematical background (**8.79 & 8.90 CGPA**) and hands-on expertise building production-ready web platforms, distributed microservices, and high-performance RESTful APIs. Proficient in Python, TypeScript, React, Next.js, FastAPI, and Docker, with a proven focus on writing clean, maintainable code, optimizing runtime latency, and deploying containerized applications. Adept at turning complex requirements into robust software solutions.

EDUCATION

Master of Science (M.Sc.) in Computer Science <i>Vidyasagar University</i> CGPA: 8.79/10	09/2023 – 07/2025 <i>Medinipur, India</i>
Bachelor of Science (B.Sc. Honors) in Mathematics <i>Vivekananda Satavarshiki Mahavidyalaya</i> CGPA: 8.90/10	07/2019 – 08/2022 <i>Manikpara, India</i>

TECHNICAL SKILLS

Languages:	Python, JavaScript (ES6+), TypeScript, SQL, HTML, CSS
Databases:	PostgreSQL, MS SQL Server, Redis
Frameworks & Libraries:	React.js, Next.js, Node.js, Tailwind CSS, FastAPI, Django,
DevOps Tools:	Docker, Git, GitHub, Linux, AWS
Core Competencies:	Data Structures & Algorithms, Object-Oriented Design, RESTful APIs

PROJECTS

CodeGen — In-Browser AI IDE | *Next.js, React, TypeScript, WebContainers, Convex* GitHub | Live | 2025

- Architected a full-stack in-browser IDE executing real Node.js runtimes client-side via WebContainer API, eliminating server compute costs and enabling zero-latency terminal commands.
- Integrated CodeMirror 6 with context-aware AI completions, AST syntax highlighting, multi-tab file navigation, and real-time state synchronization (**<50ms latency**) with Convex backend.
- Built autonomous multi-file AI coding agents capable of analyzing full repository context and executing file operations directly inside the virtual file system.

YouTube Transcript AI Assistant | *Python, LangChain, OpenAI API, Streamlit, React* GitHub | 2024

- Engineered a semantic Retrieval-Augmented Generation (RAG) platform allowing users to query long-form video lectures and podcasts with grounded responses delivered in **<1.2s query time**.
- Automated transcript extraction and cross-lingual translation pipelines supporting **50+ languages** using YouTube Transcript API and Deep-Translator.
- Implemented semantic chunking and vector search retrieval to pinpoint exact video timestamp citations, maintaining an exceptionally low hallucination rate of **<1.5%**.

Autonomous AI Web Scraper API | *FastAPI, Python, Docker, Selenium, BeautifulSoup* GitHub | 2024

- Designed an asynchronous web extraction microservice leveraging headless Selenium and BeautifulSoup to reliably scrape dynamic single-page applications (SPAs).
- Integrated LangChain reasoning agents to parse unstructured HTML into strictly validated, typed JSON schemas with **99.1% parsing accuracy**.
- Implemented asynchronous FastAPI endpoints with job queue monitoring, background tasks, and Docker multi-stage containerization for seamless cloud deployment.

Plant Leaf Disease Inference Service | *Python, PyTorch, EfficientNet-B4, FastAPI, Docker* GitHub | 2025

- Developed a computer vision diagnostics service achieving **98.17% test accuracy** classifying grapevine leaf diseases across 4 conditions using fine-tuned EfficientNet-B4 transfer learning on 9,000+ images.
- Engineered high-throughput REST API inference endpoints in FastAPI with input validation, image normalization pipelines, and **<100ms request latency**.
- Containerized the entire deep learning serving stack using Docker to ensure reproducibility across development, staging, and production environments.

ACHIEVEMENTS

- Problem Solving:** Solved **150+** algorithmic challenges on LeetCode covering array, linked list and dynamic programming.